Design and Simulation of a MOSFET-Based Widlar Current Mirror

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***Abstract*—This paper presents the design and simulation of an NMOS-based Widlar current mirror in Cadence Virtuoso using the GPDK180 technology library. The circuit uses two NMOS transistors with equal dimensions, a $100\,\mu\mathrm{A}$ reference current, a source-degeneration resistor, and a $1\,\mathrm{M}\Omega$ output load. The supply voltage is $1.8\,\mathrm{V}$, and the target output current is $1\,\mu\mathrm{A}$. A square-law MOSFET approximation gives an initial source resistance of approximately $203.7\,\mathrm{k}\Omega$. A DC parametric sweep then identifies approximately $301.8536\,\mathrm{k}\Omega$ as the resistance at which the simulated output current reaches $1\,\mu\mathrm{A}$. The difference between the hand calculation and simulation highlights the limitations of a simplified analytical model and the importance of technology-model-based verification.**

***Index Terms*—Widlar current mirror, MOSFET, current reference, analog integrated circuits, Cadence Virtuoso, GPDK180.**

I. Introduction

Current mirrors are widely used in analog integrated circuits to establish bias currents for amplifiers and other circuit blocks. In a basic MOS current mirror, a diode-connected reference transistor establishes a gate–source voltage that is applied to an output transistor. With matched devices and suitable operating conditions, the output current is related to the reference current through the transistor aspect ratios [1], [2].

A conventional mirror with equal transistor dimensions ideally reproduces approximately the same current. Generating a much smaller current can require a large device-ratio difference or additional circuitry. The Widlar principle introduces a resistor in the source of the output NMOS, creating a source voltage that reduces the output transistor's effective gate–source voltage and therefore its current. The original Widlar circuit was introduced as a low-current source using bipolar transistors; the same source-degeneration principle can also be implemented using MOSFETs [1], [3].

The objective of this assignment is to design and simulate an NMOS Widlar current mirror with a $100\,\mu\mathrm{A}$ reference current and a target output current of $1\,\mu\mathrm{A}$, then determine a source resistance that meets the target in the GPDK180 simulation.

II. Circuit Configuration and Design

A. Topology and specifications

The designed circuit contains two NMOS transistors, M1 and M2. M1 is diode-connected, and its gate is tied to the gate of M2. M1's source is grounded. M2's source is connected to ground through the source resistor R_S. A $100\,\mu\mathrm{A}$ DC current source biases the diode-connected branch, while $R_{\mathrm{Load}} = 1\,\mathrm{M}\Omega$ connects the output drain node to V_{DD}. The supply voltage is $1.8\,\mathrm{V}$. The two NMOS devices use $W = 2\,\mu\mathrm{m}$ and $L = 180\,\mathrm{nm}$.

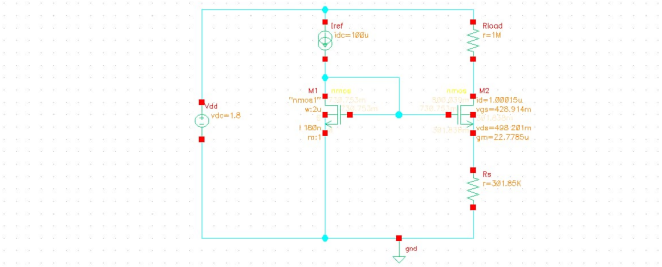


Fig. 1. Cadence Virtuoso schematic with the selected source resistance $R_S = 301.85\,\mathrm{k}\Omega$.

TABLE I
Design Specifications

Parameter	Value
Supply voltage, V_{DD}	$1.8\,\mathrm{V}$
Reference current, I_{REF}	$100\,\mu\mathrm{A}$
Target output current, I_{OUT}	$1\,\mu\mathrm{A}$
Transistor dimensions	$W = 2\,\mu\mathrm{m}$, $L = 180\,\mathrm{nm}$
Assumed V_{TH}	$0.54\,\mathrm{V}$
Assumed $\mu_n C_{\mathrm{ox}}$	$350\,\mu\mathrm{A\,V^{-2}}$
Load resistance, R_{Load}	$1\,\mathrm{M}\Omega$

The principal design values are summarized in Table I. The threshold voltage and process transconductance parameter used for the hand calculation are assumed values; they are not asserted to be the exact model parameters used internally by GPDK180.

B. Analytical estimate of the source resistance

For an NMOS transistor in saturation, the simplified square-law model, neglecting channel-length modulation, is

$$I_D = \frac{1}{2}\mu_n C_{\mathrm{ox}} \frac{W}{L}(V_{GS} - V_{TH})^2. \tag{1}$$

Rearranging (1) gives

$$V_{GS} = V_{TH} + \sqrt{\frac{2I_D}{\mu_n C_{\mathrm{ox}}(W/L)}}. \tag{2}$$

The aspect ratio is $W/L = 2/0.18 = 11.111$. With the assumed parameters, the reference-branch gate–source voltage is estimated as

$$V_{GS1} = 0.54 + \sqrt{\frac{2(100)}{350(11.111)}} \approx 0.7663\,\mathrm{V}. \tag{3}$$

For the target output current,

$$V_{GS2} = 0.54 + \sqrt{\frac{2(1)}{350(11.111)}} \approx 0.5626\,\mathrm{V}. \tag{4}$$

Assuming the same gate voltage is applied to both devices, the source voltage of M2 is approximated by $V_{GS1} - V_{GS2}$. Applying Ohm's law to R_S,

$$R_S \approx \frac{V_{GS1} - V_{GS2}}{I_{\mathrm{OUT}}} = \frac{0.7663 - 0.5626}{1 \times 10^{-6}\,\mathrm{A}} \approx 203.7\,\mathrm{k\Omega}. \tag{5}$$

This is an initial estimate only. It assumes idealized square-law behavior and does not fully model body effect, short-channel effects, subthreshold conduction, or the detailed device characteristics in the process design kit.

III. Simulation Methodology

The schematic was implemented in Cadence Virtuoso using the GPDK180 library. The DC operating point was examined, the source resistance was swept over a range of values, and a DC supply-voltage sweep and a transient analysis were also recorded. The DC resistance sweep was used to select the value of R_S that brings the output current close to the $1\,\mu\mathrm{A}$ target. The $1\,\mathrm{M\Omega}$ load resistor was retained for the simulations shown.

IV. Results and Discussion

A. Operating point

At the selected resistance, the schematic annotations report I_D for M2 of approximately $1.000\,15\,\mu\mathrm{A}$, $V_{GS2} \approx 428.914\,\mathrm{mV}$, and $V_{DS2} \approx 498.201\,\mathrm{mV}$. The reported source-node voltage is approximately $301.838\,\mathrm{mV}$. These values are the simulator's annotated operating-point results. In particular, the simulated V_{GS2} differs from the hand estimate because the analytical calculation uses assumed device parameters rather than the complete GPDK180 transistor model.

B. Source-resistance sweep

Fig. 2 shows the simulated output current as R_S varies from approximately $180\,\mathrm{k\Omega}$ to $350\,\mathrm{k\Omega}$. The current falls monotonically as the resistance increases, consistent with the source-degeneration mechanism. The plot marker identifies $R_S \approx 301.8536\,\mathrm{k\Omega}$ at the $1\,\mu\mathrm{A}$ reference line.

The transient-sweep legend provides the following current values for selected resistances; these values are also useful for cross-checking the DC curve.

The table shows that $R_S = 300\,\mathrm{k\Omega}$ gives approximately $1.005\,\mu\mathrm{A}$, while the finer DC sweep selects approximately $301.8536\,\mathrm{k\Omega}$ for the $1\,\mu\mathrm{A}$ target. The selected resistance is about 48.2% greater than the analytical estimate in (4). This discrepancy is not removed by adjusting the reported values; it reflects the difference between the simplified hand model and the technology-model simulation.

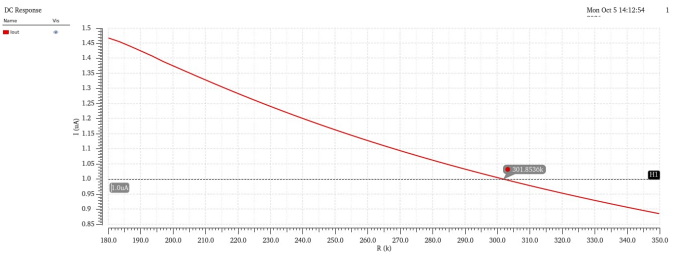


Fig. 2. DC sweep of output current versus source resistance.

TABLE II
Output Current at Selected Source Resistances

R_S (kΩ)	I_{OUT} (µA)
200	1.376448
225	1.262163
250	1.163199
275	1.078284
300	1.005004
325	0.9412397
350	0.8852925

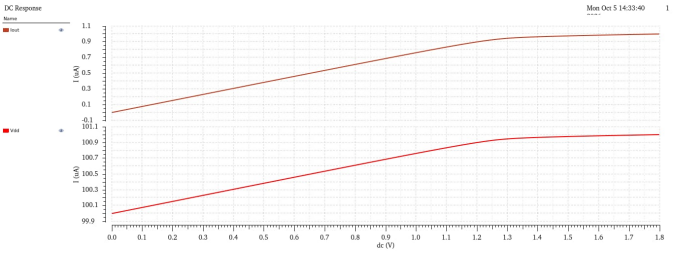


Fig. 3. DC response during the supply-voltage sweep; upper trace is I_{OUT} and lower trace is the current associated with the supply source as labelled in the simulator.

C. Supply-voltage sweep

Fig. 3 presents the recorded DC response while the supply voltage is swept from $0\,\mathrm{V}$ to $1.8\,\mathrm{V}$. The output current increases with supply voltage and approaches approximately $1\,\mu\mathrm{A}$ near the upper end of the sweep. The reference current remains close to $100\,\mu\mathrm{A}$, increasing slightly in the displayed plot. This result indicates that the output branch needs sufficient supply headroom to establish the intended current; the output current is not constant at very low supply voltage.

D. Transient plot

Fig. 4 shows the recorded transient analysis for the selected source-resistance values. The plotted currents are nearly horizontal over the displayed time interval, indicating that the recorded values remain approximately steady during this analysis. Since the plot compares different resistor settings rather than showing a clearly defined start-up stimulus, it should be interpreted as a steady-current comparison across the selected parameter values, not as a measurement of start-up settling time.

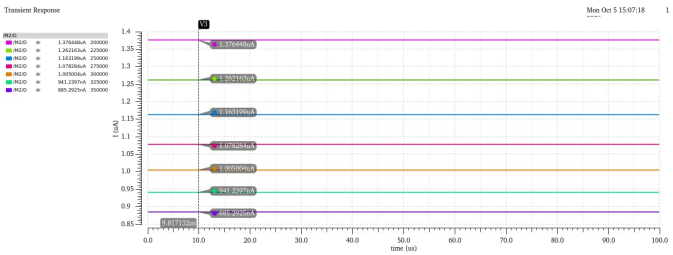


Fig. 4. Recorded transient response for several source-resistance values.

V. Advantages, Limitations, and Applications

The source resistor provides a simple means of reducing the output current without changing the two transistors' nominal aspect ratios. This is useful in low-current biasing networks and analog integrated circuits [1], [2]. The topology is relatively simple, but its accuracy depends on the transistor model, resistor value, output voltage, temperature, and device mismatch. The source resistor also consumes voltage headroom, and the output current can depart from its target when the output transistor does not have adequate drain voltage. For integrated implementation, resistor area and process variation may also be important design considerations.

VI. Conclusion

An NMOS Widlar current mirror was designed in Cadence Virtuoso with the GPDK180 library to produce approximately $1\,\mu\mathrm{A}$ from a $100\,\mu\mathrm{A}$ reference current at a $1.8\,\mathrm{V}$ supply. The square-law hand calculation gave $R_S \approx 203.7\,\mathrm{k}\Omega$, while the source-resistance sweep identified $R_S \approx 301.8536\,\mathrm{k}\Omega$ for an output current close to the target. The plots confirm that increasing the source resistance reduces the output current and that the output current depends on the available supply voltage. The exercise demonstrates the value of analytical estimates followed by simulation with the intended technology models.

References

[1] R. J. Widlar, "Low-Value Current Source for Integrated Circuits," U.S. Patent 3,320,439, May 16, 1967. [Online]. Available: https://patents.google.com/patent/US3320439A/en

[2] Analog Devices, "Chapter 11: The Current Mirror," Analog Devices Wiki, section on the Widlar current source. [Online]. Available: https://wiki.analog.com/university/courses/electronics/text/chapter-11

[3] P. R. Gray, P. J. Hurst, S. H. Lewis, and R. G. Meyer, *Analysis and Design of Analog Integrated Circuits*, 5th ed. Hoboken, NJ, USA: Wiley, 2009.